Alapan Das

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Professional Summary

B.Tech in IT (CGPA 8.6) with a peer-reviewed publication and 6th rank in ACM UG Project Contest 2026. Built real life problem solving production-grade ML systems & Data Center Solutions. Excel at Computer Networks and Topologies. Aiming for an SDE/AI-ML role to apply strong CS fundamentals in networking, systems, and applied ML.

Technical Skills

Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks
Languages: Python, C++, Java, C, SQL
AI / ML: PyTorch, OpenCV, TensorFlow, TorchVision, CNNs, Computer Vision
Networking: Mininet, Open vSwitch, OpenFlow, SDN, TCP/IP, NetworkX
Backend & Web: Flask, Flask-SocketIO, Node.js, HTML, CSS, JavaScript
Databases: MySQL, Oracle, MongoDB
Tools & Platforms: Git, GitHub, Docker, Vercel

Experience

IBM SkillsBuild Internship June - Aug'24

- Developed responsive frontend components for FoodWise with **HTML, CSS, JavaScript** & integrated **RESTful APIs**.
- Used **GitHub** for version control & development. Contributed to the scalability of web application in **Agile** environment.

Projects

Hybrid Routing for Data Centers (Documentation) ***(Python, Mininet, Open vSwitch, OpenFlow)***

- Designed hybrid Leaf-Spine + De Bruijn topology with deterministic arithmetic flowlet routing replacing ECMP hashing.
- Architecture validated in Mininet. **258K+ datagrams - 0% ping loss**, **0% UDP loss**, **38.7 Gbps peak over 120s** under simulated load.

Smart Traffic Management System (Github) ***(Python, Flask, PyTorch, OpenCV, NetworkX)***

- Fine-tuned ResNet-50 CNN. Classified 3 live traffic densities - **97% precision**, **90% recall** on Heavy Traffic over **279 test samples**.
- Built an adaptive signal-timing and route-optimization engine using REST APIs, WebSockets, NetworkX to reroute traffic dynamically based on real-time congestion data.

Driver's Drowsiness Alert System (Github) ***(Python, Flask, OpenCV, Dlib)***

- Real-time detection pipeline with OpenCV and Dlib - **17 FPS** in **low light**, robust upon wearing **spectacles/ helmets**.
- Facial landmark detection & eye-blink-ratio analysis - classifies driver state at **91% accuracy**, **89% precision**, **0.2s voice-alert latency**.

Publications

"Exploring the Potential of De Bruijn Graph Topology to Serve Next-Generation Data Centres"- peer-reviewed & accepted at the 13th International Symposium on Applied Computing for Software and Smart Systems and will be published in the Springer Nature Series. July 2026

Achievements

Special mention for Research paper at ACSS 2026 July 2026
6th position in ACM Kolkata project contest May 2026
Smart Bengal Hackathon Finalist May 2025

Education

B.Tech in Information Technology | *University of Calcutta* | CGPA: 8.60 2023-2026
B.Sc in Computer Science (Hons.) | *University of Calcutta* | CGPA: 8.37 2020-2023